

Assignment 5

1. How do you interpret the results of these analyses?

Figure 1 and Figure 2 show wordclouds from the Apple Corpus; the more the word is used, the larger that word is in the cloud. Stop words, and the words “apple”, “iphone”, and “ipad”, were removed from the wordclouds. The additional words were removed because they are the keywords used to create the corpus and are expected in large quantity. With filtering, the largest words in these figures are “pro”, “15”, “phone”, “max”, “android”, “amp”, “16”, “samsung”, “people”, and “camera”. Additionally, I searched the corpus to see the context of these most used words. Below are examples of the posts; further analysis was not conducted as none of the posts seemed notably significant:

- “@Apple @tim_cook shame on you again, iphone 16 pro max sam,e shedy desig taha 4 years ago, iphone 16 sema shedy design tahs 6 ayres ago @cnni @FoxNews @SamsungMobile @Huawei @pixels_online the worst part of it, i was iphone user until now, if iphoe 16 pro max it's = gonna change”
- “Iphone vs Samsung”
- “Suitable for iPhone 15, 14, 13, 12 Pro Max Plus Mini Megsafe Magnetic Card Holder PU Leather Back Sticker Card Case Phone Leather Wallet”
- “iPhone problems”
- “I just tried Apple Intelligence on my iPhone 15 Pro — here's what I found out | Tom's Guide...”

Figure 3 shows the plot of the mean sentiment of the entire corpus, with dates from 2022 to August 1, 2024. As seen in Figure 3, the range is so large that the cluster of posts in July/August 2024 are too close together and cannot be accurately analyzed. Because of this, the dates prior to July 28th were removed and the graphs were replotted as seen in Figures 4 -6, ranging from July 28, 2024, to August 1, 2024.

Within these Figures 4-6, July 28 and July 29 have a mean sentiment close to zero. The mean sentiment increases on July 30 and July 31, with the highest positive mean sentiment on July 31. The mean sentiment then declines and returns to a mean sentiment close to zero.

The last graph in Figure 6 is an area chart of the mean sentiments from July 28, 2024, to August 1, 2024. This graph shows a mean negative sentiment on July 28 close to zero, with a decline on July 29, and at its highest negative mean sentiment on July 30. On July 31, the mean sentiment dramatically shifts in the positive, and on August 1, the mean sentiment is negative, but closer to the baseline of zero.

2. What are the business implications of each visualization?

There is an implication that the tweets are likely discussing Apple products and its competitor, Samsung. Additionally, different Apple products and their features, including Apple Intelligence are discussed. The wordclouds are visually appealing, but I found the Topic Models and Text Networks to summarize the corpus best. If I were to do a similar analysis for work, I would ensure to include multiple variations of analysis to get a wholistic understanding.

The mean sentiment graphs indicate a significant change in sentiment on July 30 and July 31. Figures 4 and 5 indicate a positive trending mean sentiment over those two days. However, the area plot in Figure 6, shows steep negative mean sentiment on July 30, and a steep positive mean sentiment on July 31. This is not a surprise, as the area chart shows the extreme cases in the corpus, and the scatter plots show the mean sentiment of the corpus on a whole. There is an overall positive mean sentiment, which could indicate Apple released or announced a product or feature that users are excited about and posting to X in a positive manner. This data could be used to

make operational and business decisions in the future that encourage an overall positive sentiment around Apple products from users.

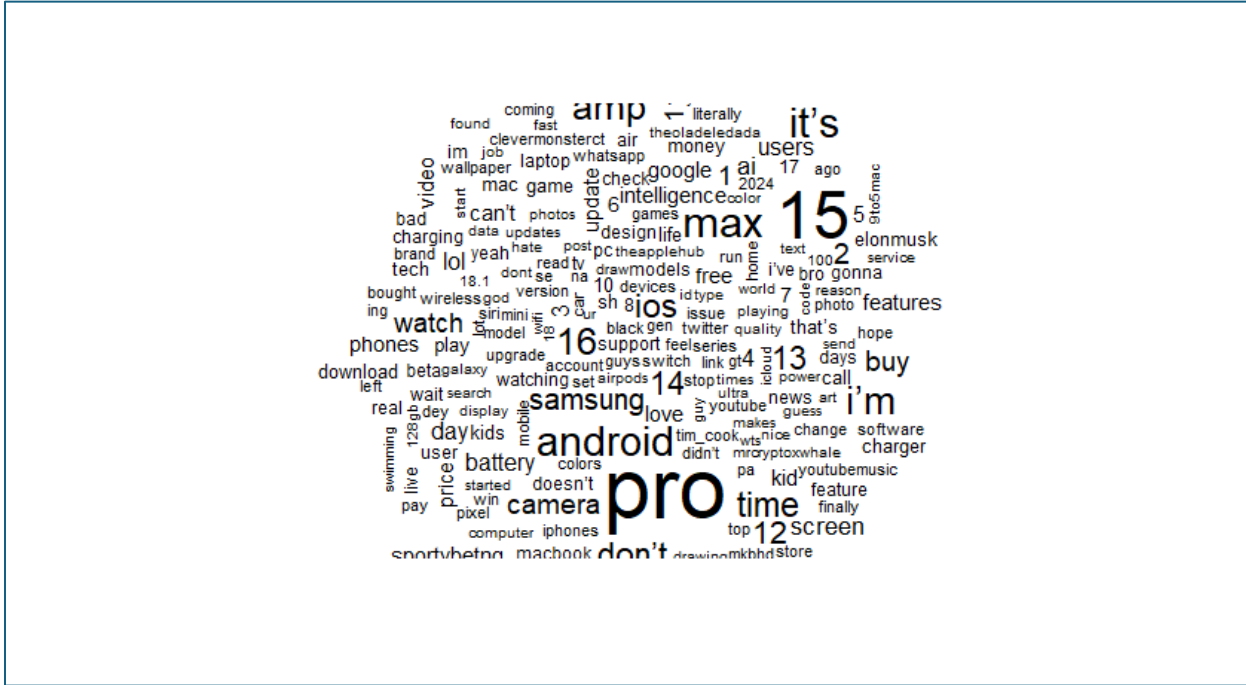


Figure 1 Basic Word Cloud with wordcloud package.

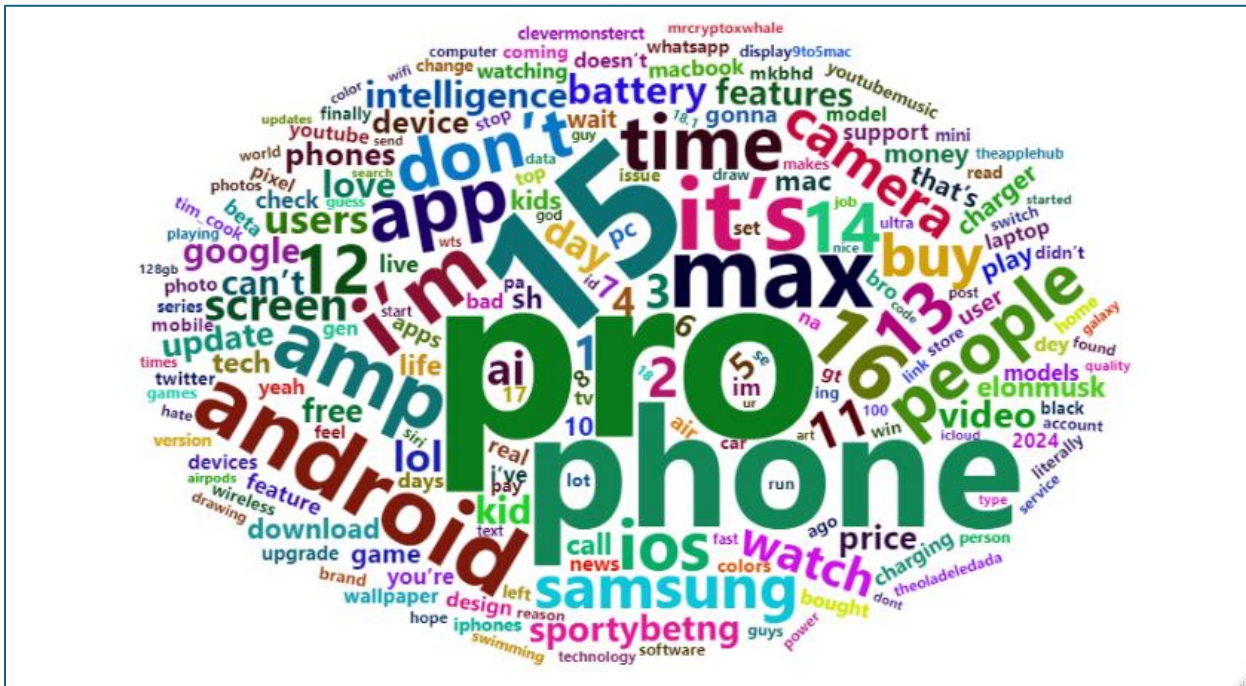


Figure 2 Wordcloud with wordcloud2 package.

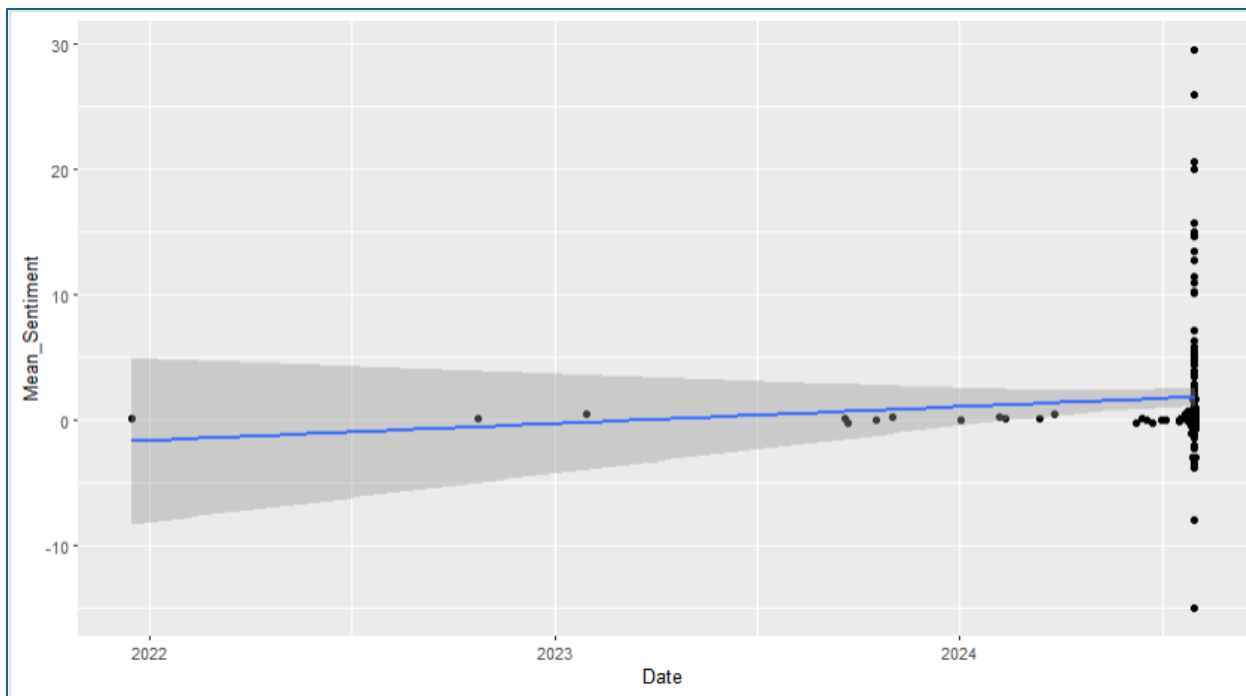


Figure 3 Mean Sentiment Over Time without Date Filtering

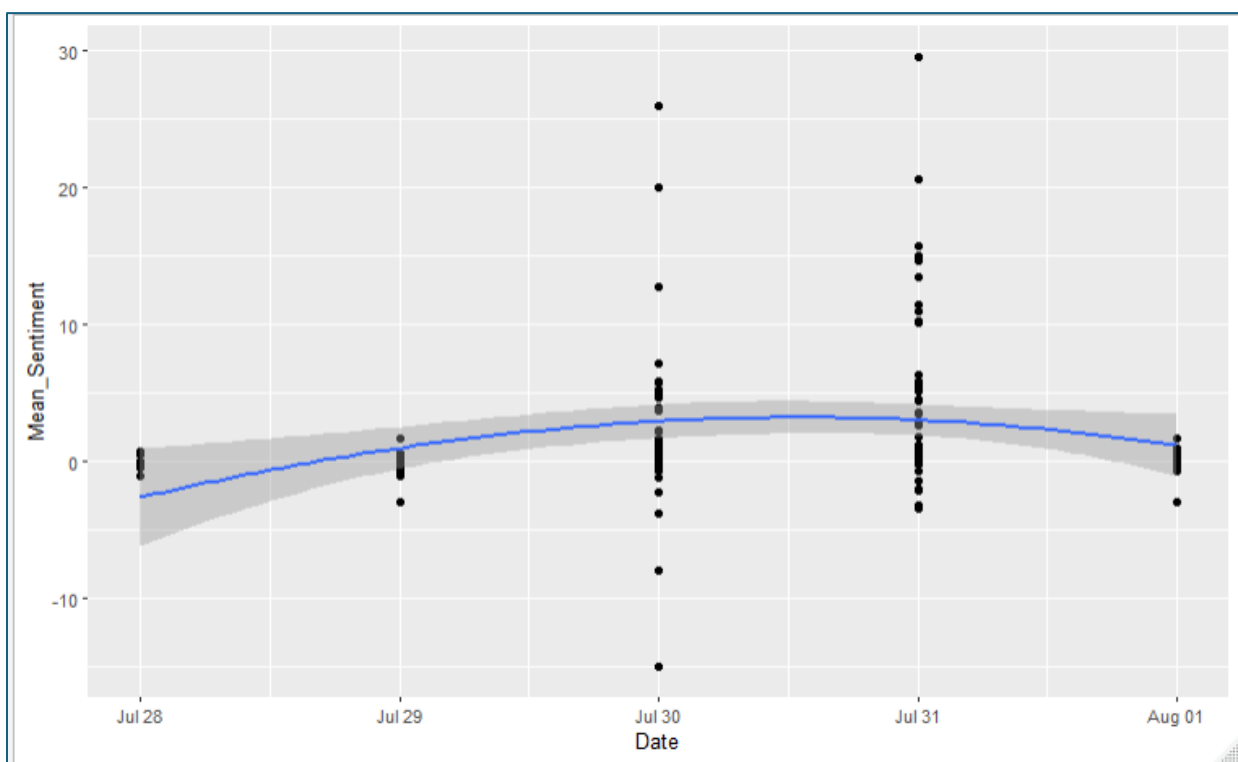


Figure 4 Mean Sentiment Over Time (weighted w. uncertainty and smooth trend line) from 7-28-24 to 8-1-24.

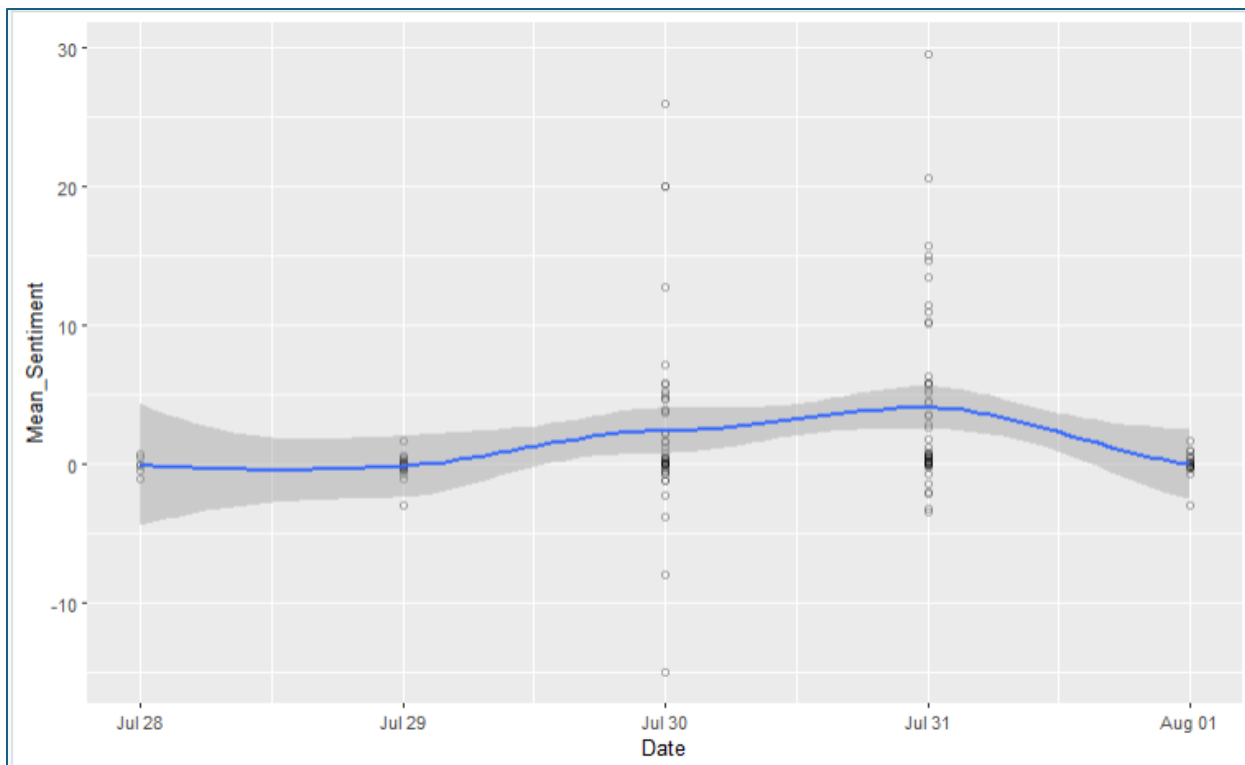


Figure 5 Mean Sentiment Over Time from 7-28-24 to 8-1-24 (Final Adjustments).

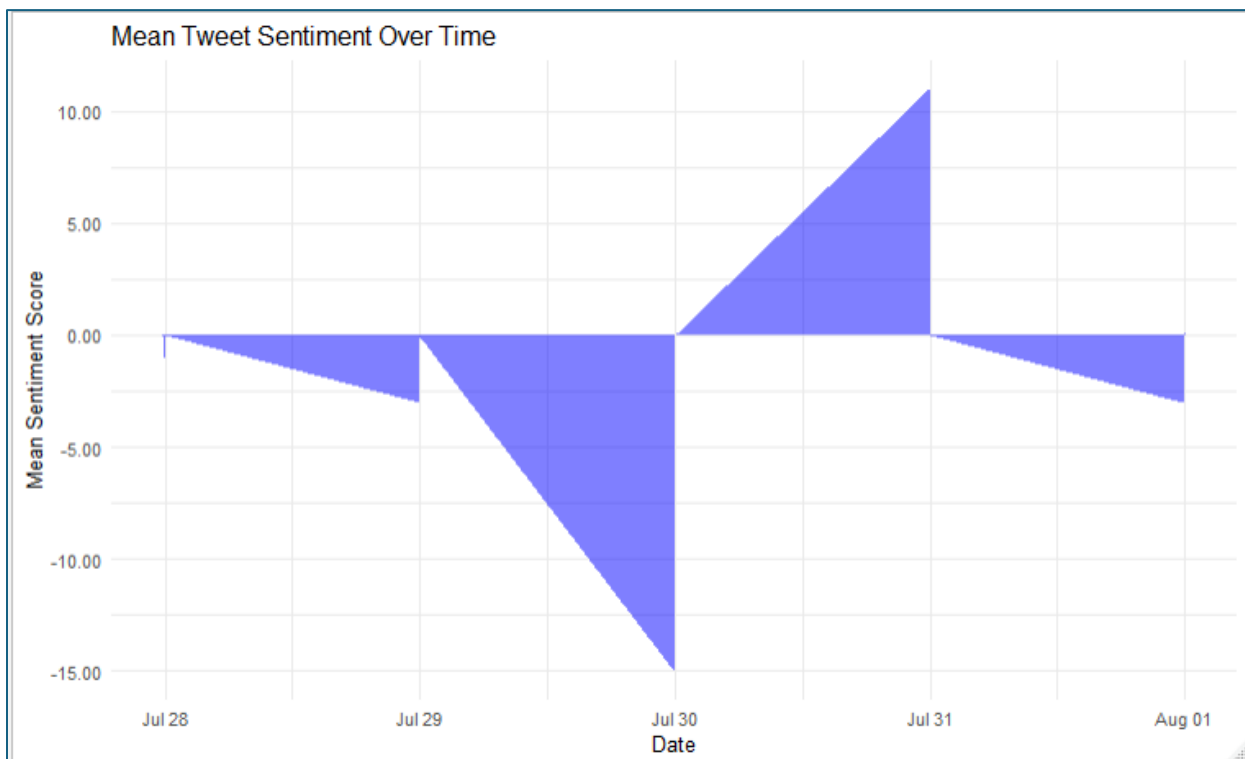


Figure 6 Area Chart of Mean Sentiments from 7-28-24 to 8-1-24.